

EXPERIMENT – 7

AIM

To write a program to implement flow control at the data link layer using the **Sliding Window Protocol** and simulate the flow of frames from one node to another.

THEORY

The Sliding Window Protocol is a flow-control technique used at the data link layer. It allows the sender to transmit multiple frames before waiting for an acknowledgement from the receiver.

In this experiment, each character of the input message is considered as one frame. Each frame contains a **Frame Number** and **Data**. The sender transmits frames according to the specified window size.

The receiver reads the frames from the Sender Buffer and checks whether the frame numbers are received in the expected order. If the frame is correct, the receiver generates an **ACK**. If an incorrect frame is received, a **NACK** is generated.

The sender reads the acknowledgement from the Receiver Buffer. If the expected ACK is received, the sender moves the window forward and transmits the next set of frames. If a NACK is received, the required frames are retransmitted.

The experiment also allows errors to be introduced manually by changing the frame number or acknowledgement number in the corresponding files, as specified in the lab manual.

ALGORITHM

Sender

1. Start the program.
2. Read the window size from the user.
3. Read the text message from the user.
4. Divide the message into frames, considering one character as one frame.
5. Assign a frame number to every character.
6. Create the frames containing Frame Number and Data.
7. Send the frames according to the window size.
8. Store the transmitted frames in Sender_Buffer.
9. Wait for the acknowledgement from the receiver.
10. Read the Receiver_Buffer.
11. Check the acknowledgement number.
12. If the expected ACK is received, move the window forward.
13. Send the next set of frames.
14. If NACK is received, retransmit the required frame(s).

15. Continue until all frames are successfully transmitted.

16. Stop.

Receiver

1. Start the receiver.
2. Read the frames from Sender_Buffer.
3. Check the frame number.
4. Compare it with the expected frame number.
5. If the frame number is correct, accept the frame and generate an ACK.
6. If the frame number is incorrect, generate a NACK.
7. Store the acknowledgement in Receiver_Buffer.
8. Continue until all frames are received.
9. Stop.

PROGRAM

```
import time
```

```
message = input("Enter the message: ")
```

```
window = int(input("Enter window size: "))
```

```
frames = []
```

```
for i, ch in enumerate(message):
```

```
    frames.append([i + 1, ch])
```

```
i = 0
```

```
while i < len(frames):
```

```
    end = min(i + window, len(frames))
```

```
    print("\nSending frames:")
```

```
with open("Sender_Buffer", "w") as f:
    for j in range(i, end):
        print("Frame No:", frames[j][0], "Data:", frames[j][1])
        f.write(str(frames[j][0]) + " " + frames[j][1] + "\n")
```

```
time.sleep(1)
```

```
expected = i + 1
```

```
with open("Receiver_Buffer", "w") as f:
    for j in range(i, end):
        if frames[j][0] == expected:
            f.write("ACK " + str(frames[j][0]) + "\n")
            print("Receiver: ACK", frames[j][0])
            expected += 1
        else:
            f.write("NACK " + str(expected) + "\n")
            print("Receiver: NACK", expected)
            break
```

```
with open("Receiver_Buffer", "r") as f:
```

```
    ack = f.readlines()
```

```
success = True
```

```
for line in ack:
```

```
    parts = line.split()
```

```
    if parts[0] == "NACK":
```

```
        success = False
```

```
        break
```

```
if success:
```

```
    i = end
```

```
else:
```

```
    print("Retransmitting frames...")
```

```
    time.sleep(1)
```

```
print("\nAll frames transmitted successfully.")
```

```
print("Message:", message)
```

SAMPLE INPUT

Enter the message: HELLO

Enter window size: 3

SAMPLE OUTPUT

Sending frames:

Frame No: 1 Data: H

Frame No: 2 Data: E

Frame No: 3 Data: L

Receiver: ACK 1

Receiver: ACK 2

Receiver: ACK 3

Sending frames:

Frame No: 4 Data: L

Frame No: 5 Data: O

Receiver: ACK 4

Receiver: ACK 5

All frames transmitted successfully.

Message: HELLO

OBSERVATION

Frame No. Data Acknowledgement

1	H	ACK 1
2	E	ACK 2
3	L	ACK 3
4	L	ACK 4
5	O	ACK 5

For a window size of 3, the sender initially transmits three frames. After receiving the acknowledgements, the window moves forward and the next frames are transmitted.

If an incorrect frame number is introduced, the receiver generates a NACK and the sender retransmits the required frames.

RESULT

Thus, the **Sliding Window Protocol** was successfully implemented and the flow of frames between the sender and receiver was simulated using acknowledgements and retransmissions.